

Supplementary Note 1

PAC formalism for resistance learnability

Companion to: N_e^ : a computable population-size threshold for resistance learnability from standard tumour sequencing*

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This note summarizes the learning-theoretic objects behind N_e^* for reviewers of the empirical validation paper. Full theorem statements and proofs are not reproduced here; they appear in the companion theory preprint:

Stillwell, R. C. Natural selection is empirical risk minimisation. Preprint at Research Square (2026). <https://doi.org/10.21203/rs.3.rs-9848110/v1>

That preprint establishes the unified empirical-risk-minimization structure linking Fisher's theorem, the Price equation, and PAC sample-complexity bounds. The empirical paper tests predictions derived from that framework without duplicating its proofs.

Instance space and hypothesis class

Instance space X : bulk-sequencing summaries of resistance-relevant genotype structure at baseline—allele frequencies at L independently curated driver loci, together with subclonal dispersion V_A across the genome.

Hypothesis class H : resistance phenotypes realizable by selection on a tumour effective population of size N_e with subclonal variance V_A . Each h in H specifies a distribution over resistant clone frequencies achievable within one evolutionary episode before sensitive cells are depleted.

Target concept h^* : an ε -optimal resistant phenotype—fitness within ε of the best resistance genotype discoverable at the L loci under the current therapy.

Loss and failure event

Loss (per evolutionary episode):

$$\ell(h) = \max(0, f(h^*) - f(h)),$$

where $f(\cdot)$ is resistance fitness under drug pressure, normalized so $f(h^*) = 1$.

Failure event (PAC sense): $\ell(h) > \varepsilon$ after search over N_e inherited units—i.e. evolution does not reach an ε -optimal resistant state with the required reliability.

This is a continuous, Fisher-information learning problem on allele-frequency space, not binary classification. VC dimension of threshold functions on R^d is therefore not the operative capacity measure.

Role of N_e

N_e enters as effective sample size: the number of independently segregating units that contribute to sampling resistance alleles before selection exhausts sensitive clones. The bound is

$$\epsilon^*(N_e, V_A, L) = \sqrt{2 V_A [L \ln(e N_e / L) + \ln(4/\delta)] / N_e},$$

where the tail $P(\text{failure}) \leq \delta$ follows a large-deviation (Chernoff–Hoeffding type) inequality on occupancy of resistance alleles across L loci, with V_A scaling subclonal noise. N_e is sample complexity in this stochastic search sense, analogous to m in PAC sample-size bounds but not to labeled-example count in Valiant's discrete hypothesis classes. See Stillwell (2026 preprint) for derivation under the infinitesimal model.

Threshold N_e^*

N_e^* is the smallest N_e such that $P(\text{failure}) \leq \delta$ at tolerance ϵ , obtained by monotone bisection on the PAC tail (implementation: `threshold.ne_star` in the analysis repository).

Relation to empirical analyses

Cross-cancer ecology (Fig. 2 of the empirical paper): tests whether $\epsilon^*(N_e, V_A, L)$ covaries with published resistance kinetics when L is fixed independently. This does not require PAC identifiability to hold pointwise for every tumour.

Patient analyses: test whether $\log_{10}(N_e/N_e^*)$ stratifies time-to-resistance when inputs are estimated from single samples.